WORKED SOLUTION

STEP III 1999, Question 11

Mechanics • Circular Motion

STEP III 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q11

The key idea

Once the hoop is on the edge it is a rigid body pivoting about a fixed point, so two equations do all the work: conservation of energy gives the angular speed at any angle, and Newton's second law for the centre, resolved towards the pivot, gives the normal reaction. Contact is lost when that reaction falls to zero. The moment of inertia from the first part is exactly what the energy equation needs.

Solution

The moment of inertia

Every particle of a thin hoop is at distance a from its centre C, so about the axis through C perpendicular to the plane of the hoop the moment of inertia is

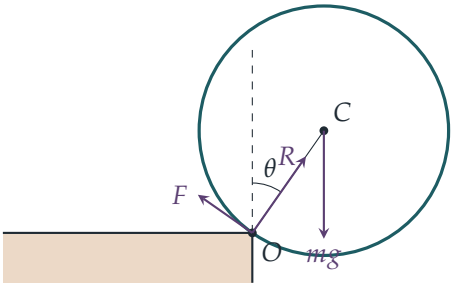
$$I_C = \sum \delta m\, a^2 = ma^2.$$

A point O on the circumference is at distance a from C, and the new axis is parallel to the old one, so by the parallel axis theorem

$$I_O = I_C + ma^2 = 2ma^2.$$

Rolling over the edge

Model the hoop as a uniform rigid ring and the edge of the table as a fixed point O. Both surfaces are rough, and we assume friction is large enough that the hoop never slips: on the table it rolls, and over the edge it turns about O without sliding. Let θ be the angle that OC makes with the upward vertical, so $\theta = 0$ at the moment the hoop reaches the edge, and write $\omega = \dot{\theta}$.



Ask yourself What is the hoop's angular speed about O at the instant it reaches the edge?

A hoop rolling without slipping at speed v has angular speed v/a, and its motion at each instant is a rotation about its point of contact. At the instant it reaches the edge, the point of contact *is* O, and the hoop simply carries on turning about it: no impact occurs, since the hoop does not lose contact. So $\omega = v/a$ when $\theta = 0$.

Energy. While the hoop stays in contact with O, the forces at O do no work (the point of application is at rest), so mechanical energy is conserved. The centre has dropped by $a - a\cos\theta$, so

$$\tfrac{1}{2}(2ma^2)\,\omega^2 = \tfrac{1}{2}(2ma^2)\left(\frac{v}{a}\right)^2 + mga(1-\cos\theta), \qquad \text{that is} \qquad a\omega^2 = \frac{v^2}{a} + g(1-\cos\theta).$$

Towards the pivot. The centre C moves on a circle of radius a about O, so its acceleration has component $a\omega^2$ directed from C towards O. The forces on the hoop are its weight mg, the normal reaction R along OC (pushing away from O), and friction F perpendicular to OC, which has no component along OC. Resolving along CO:

$$mg\cos\theta - R = ma\omega^2.$$

Substituting the energy equation,

$$R = mg\cos\theta - \frac{mv^2}{a} - mg(1-\cos\theta) = mg(2\cos\theta - 1) - \frac{mv^2}{a}.$$

This decreases as θ increases, and the hoop loses contact with the edge when R first reaches zero, which happens when

$$2g\cos\theta = g + \frac{v^2}{a}, \qquad \text{that is} \qquad \cos\theta = \frac{1}{2} + \frac{v^2}{2ag},$$

as required.

Ask yourself Does this make sense for every v?

At $\theta = 0$ we have $R = m(g - v^2/a)$, which is positive only if $v^2 < ag$. That is the condition for the hoop to roll over the edge without immediately losing contact, which

the question assumes, and it is exactly the condition for the right-hand side to be less than 1, so a valid angle θ exists.

For completeness, resolving perpendicular to OC with $I_O\ddot{\theta} = mga\sin\theta$ gives $\ddot{\theta} = g\sin\theta/(2a)$ and $F = \frac{1}{2}mg\sin\theta$. This is the friction the edge must supply to prevent slipping, which is why both surfaces need to be rough.

Checks

- If $v \to 0$ the hoop just topples off the edge and loses contact at $\cos\theta = \frac{1}{2}$, so $\theta = 60°$. As $v^2 \to ag$, $\theta \to 0$: the hoop barely touches the edge. Both limits make sense.
- Numerically, with $a = 0.6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $v = 1.5\,\mathrm{m\,s^{-1}}$, stepping $\ddot{\theta} = g\sin\theta/(2a)$ from $\dot{\theta} = v/a$ until $g\cos\theta = a\dot{\theta}^2$ gives $\cos\theta \approx 0.69132$, matching $\frac{1}{2} + v^2/(2ag) = 0.69133$.

Common slips

- Using $I_C = ma^2$ in the energy equation, or adding a separate translational kinetic energy term on top of $\frac{1}{2}I_O\omega^2$. The rotation about O already includes the motion of the centre.
- Forgetting the initial kinetic energy, which gives $\cos\theta = \frac{1}{2}$ for every v.

Beyond STEP

Rolling off the edge is a typical *unilateral constraint*: the edge can push but not pull, so the equations of motion apply only while the reaction they predict is non-negative. The same idea governs a particle sliding off a sphere and a car going over a hump-backed bridge. In a first-year course you meet it alongside rigid-body dynamics and the parallel and perpendicular axis theorems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats all of these carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.